

A Comprehensive Deep Learning Approach for Deepfake Detection on the Celeb-DF Dataset

Muhammad Ubaid Ullah
Faculty of Information and
Communication Technology (FICT)
Universiti Tunku Abdul Rahman
(UTAR)
Kampar, Perak, Malaysia
ubaid.ullah@utar.my

Ming Lee Gan
Faculty of Information and
Communication Technology (FICT)
Universiti Tunku Abdul Rahman
(UTAR)
Kampar, Perak, Malaysia
ganml@utar.edu.my

Hock Guan Goh
Faculty of Information and
Communication Technology (FICT)
Universiti Tunku Abdul Rahman
(UTAR)
Kampar, Perak, Malaysia
gohhg@utar.edu.my

Aun Yichiet
Faculty of Information and
Communication Technology (FICT)
Universiti Tunku Abdul Rahman
(UTAR)
Kampar, Perak, Malaysia
aunyc@utar.edu.my

Vasaki Ponnusamy
Higher Colleges of Technology
Fujairah, UAE
vasaki.ponnusamy@gmail.com

Abstract— The rapid growth of deepfake technology has raised critical concerns regarding misinformation, identity theft, and cybersecurity risks. Detecting these manipulated media requires sophisticated algorithms capable of effectively distinguishing between real and fake content. This study provides a comparative analysis of seven convolutional neural network (CNN) architectures—ResNet50, VGG16, VGG19, Xception, InceptionV3, MobileNetV2, and MobileNetV1—using the challenging Celeb-DF v2 dataset for deepfake detection. While CNNs have been widely studied for deepfake detection, this work focuses on the trade-offs between model accuracy and computational efficiency, offering insights into how lightweight models can be optimized for real-world applications. Performance was assessed using key metrics such as accuracy, precision, recall, loss, ROC-AUC, and PR-AUC. Among the models tested, MobileNetV1 achieved the best performance, with an accuracy of 98.24% and perfect scores of 1.000 for both ROC-AUC and PR-AUC. MobileNetV1 also demonstrated a significantly lower loss of 0.0535, outperforming other models in error minimization during training. These findings provide valuable guidance for selecting efficient yet high-performing CNN models in deepfake detection, particularly in environments where computational resources are limited.

Keywords—Deepfake, Convolutional Neural Networks, MobileNet, Generative Adversarial Networks

I. INTRODUCTION

Deepfake technology is based on artificial intelligence (AI) and deep learning algorithms such as Generative Adversarial Networks (GANs), and is one of the most discussed technologies in recent times. It allows constructing hyper-realistic artificial images, videos and audio, which are challenging to differentiate with real ones. GANs are neural networks containing two neural networks: one producing the content and the other differentiating between the content and the real media have achieved great progress in the production of highly realistic synthetic media. It has become more open source technology and now has given new opportunities in the entertainment and creative sectors, i.e., virtual actors and interactive media. Nevertheless, with the development of the deepfake technology, the possibility of its abuse has become a critical issue of concern when it comes to privacy, security, and the quality of digital media. Due to the popularity of deepfake generation software, there are various abuses of the

technology. Deepfakes have been used in political disinformation campaigns as the most pressing problem, where manipulated media is used to present false information about people or false news, which may affect elections and opinion of the population [1]. What makes these videos go viral is that they create confusion and cast their doubts on the originality of digital content. Cyberbullying is another area of involvement of deepfake technology, whereby defamatory information is transmitted with the help of manipulated videos and audio. This may lead to serious emotional damage and reputation loss [2]. Also, scams have also exploited deepfake technology, such as by impersonating executives in a video or voice call to lure employees into sharing sensitive information, which has led to financial loss and security breaches [3].



Fig. 1. Real Vs Fake

The comparison shown in Fig 1 underscores the difficulty of detecting deepfakes using human observation alone, as the differences between real and fake images can be subtle and often undetectable at first glance. These inconsistencies, which may include anomalies in facial expressions, lighting, or even the alignment of facial features, are critical for deepfake detection systems to identify. The ability of deep learning models, particularly CNNs, to detect these minute discrepancies is what sets them apart from traditional methods, which often fail to identify such hidden artifacts. This highlights the importance of continued research in developing more sophisticated models capable of recognizing these subtle flaws in synthetic media, ensuring that detection technologies stay ahead of evolving deepfake generation techniques [2].

Deepfakes are not only a threat to cybersecurity but also a major issue with regard to personal harassment and misinformation. With progress in the technology of deepfake generating systems, it is becoming harder to use digital media with trust, particularly when it comes to such a situation as a video meeting or a digital transaction [4]. Deepfake technology allows a person to imitate the voice or the face of a person in a convincing way and can be used by cybercriminals to make individuals share sensitive data, which can be seen as problematic in terms of the safety of online communication and money transfer via the Internet. Conventional methods of detection, including examination of image metadata or the presence of compression artifacts, have also been shown to be ineffective against quality deepfakes [5]. The most recent state-of-the-art in deep learning, specifically using Convolutional Neural Networks (CNNs) has rendered the possible detection of subtle inconsistencies in deepfake media that cannot be detected by the naked eye. CNNs are capable of automatically extracting high-level representations on data and recognizing complex patterns, e.g. irregular facial movements and irregular lighting [6],[7]. Transfer learning has countered this, however, by demonstrating itself to be an important way of deepfake detection, where deepfake narrow datasets can be fine-tuned on large datasets such as ImageNet. It enhances the detection rate and saves the training time, particularly when the labeled data is limited, and enables models to be adapted to small visual artifacts associated with new methods of deepfakes generation [5], [8].

Although the progress of deepfake detection with CNNs has been achieved, there are still gaps in literature. Most of the research work devotes effort to attain model accuracy, thus there exists no such review to assess the trade-offs between the computation efficiency and performance that is important in the real-world application. Moreover, despite the success of CNNs, such as ResNet and VGG16, not many studies have critically evaluated the differences in the relative performance of different CNN architectures on different data sets. Also, the models usually train on the particular properties of the data, which restricts their generalizing capabilities and makes them unable to succeed on new unexpected deepfakes. [9]. This paper aims to address these gaps through the following contributions:

- Comparative Evaluation of CNN Architectures
- Trade-offs Between Accuracy and Efficiency
- Address Dataset Challenges and Overfitting Using Transfer Learning

II. METHODOLOGY

The proposed system of deepfake detection is structured as a pipeline of steps systematically preparing and pre-processing data before deep learning-based classification. It starts by decomposing video, in which samples of the Celeb-DF V2 are broken down into separate frames. To achieve a sound evaluation, redundant frames are removed and dataset is balanced to have a proportional distribution of authentic and manipulated samples in training, validation and testing subsets. Every single frame that is extracted is preprocessed and this procedure includes face detection, cropping, resizing and normalization so that models concentrate on facial

artifacts associated with manipulation instead of background noise. This is done to decrease computational work and to increase the quality of input data. The fined frames are subsequently fed into a collection of state-of-the-art convolutional neural network (CNN) models, such as ResNet50, VGG16, VGG19, Xception, InceptionV3, MobileNetV1, and MobileNetV2. We pre-train them using ImageNet weights, and they are further trained to perform binary classification between real and fake images, using transfer learning as a way to speed up convergence and generalization. The tuning of hyperparameters (e.g., batch size, learning rate, and number of epochs) during supervised training is used to obtain optimal performance, whereas regularization methods are used to reduce overfitting. It is evaluated based on several performance measures, such as accuracy precision, recall, F1-score, and area under the ROC curve (AUC).

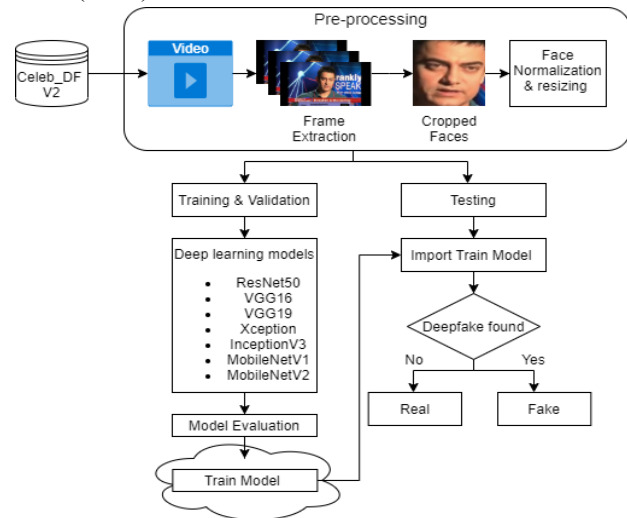


Fig. 2. Illustration of the overall workflow of the proposed method

Besides, the graphical representations like the training-validation accuracy and loss curves, ROC and precision-recall plot, bar chart comparisons and confusion matrices, present additional information on how the model behaved. The entire procedure of frame extraction to classification output is summarized in Fig 2, which shows the step by step approach of the proposed system.

A. Dataset

This paper uses the Celeb-DF v2 dataset which is one of the most difficult and realistic datasets in deepfake detection literature. Celeb-DF v2 was built to correct the weaknesses of previous datasets, which tended to have low visual quality and could have obvious artifacts like color mismatch, splicing edges, and unnatural face poses. It is composed of 590 real videos, 300 YouTube real clips, and 5639 high-quality deepfake videos that have been created with an improved synthesis process that minimizes the typical flaws of previous datasets [10],[11].

B. Pre-processing

Once the datasets were collected, some pre-processing operations were implemented to make sure the frames extracted were consistent, well-balanced, and could be trained on deep learning models. Such measures reduced the noise due to unrelated parts, handled imbalance in the classes,

and made the inputs of all samples standard. The entire process of pre-processing is as described below.

1) Frame Extraction

In the initial step, individual frames that comprised every video in the Celeb-DF v2 dataset were systematically broken down to allow a frame level analysis. Deepfake artifacts can be usually subtle and they do not necessarily appear across the whole video; instead, they can be found in short segments, or even just a frame. Frame extraction also made sure that all potential cases of manipulated content could be processed separately, which made the dataset more robust. Upon extraction, frames were separated into three subsets such as training, validation, and testing that play different roles in the process of model building and testing. Table I shows that there were 8,048 fake and 8,072 real frames in the training set, 1,772 fake and 1,761 real frames in the validation set and 1,708 fake and 1,695 real frames in the testing set. A total of 23,056 frames were harvested, split into 70 percent training, 15 percent validation and 15 percent testing, which gave both balance and representativeness.

TABLE I. FRAME SPLIT AFTER BALANCING

Set	Fake frames	Real frames	Total	Percentage
Train	8,048	8,072	16,120	69.916%
Validation	1,772	1,761	3,533	15.323%
Test	1,708	1,695	3,403	14.759%
Total			23,056	100%

2) Frame Balancing

A frame balancing strategy was used to eliminate bias towards the majority class. Most deepfake datasets are not balanced between real and fake data, and this can bias paradigms to falsely predict a particular classification. With this end in view, the number of real and fake frames was almost equal and kept in each subset. Table I shows that the distribution was almost perfectly balanced with almost no differences between real and fake samples in terms of training, validation, and testing sets. This balancing measure made sure that the models learned discriminative features, and this results in higher reliability of evaluation measures like accuracy, precision and recall.

3) Face Region Extraction

As the background factors can be excluded as the manipulations are made mostly to the face like clothing or background factors will be excluded. An efficient face detecting algorithm was used to identify and extract the face of each frame. This step minimized computation and removed irrelevant noise which compelled the models to pay attention to manipulation-specific information including texture anomalies, blending artifacts and inconsistencies in lighting. The presence of these cleaner inputs and higher detection rates was achieved by isolating the face and making the dataset cleaner.

4) Face Normalization and Resizing

Lastly, cropped faces were standardized and resized to make all CNN architectures receive standardized inputs. Light variations, camera quality and contrast differences were smoothed by equalizing pixel intensity values which is normally within the interval [0,1]. Faces were then cropped and down sampled to 224x224 pixels, matching the dimension of other architectures like ResNet, VGG and

MobileNet. This standardization increased the reproducibility of models, speeded up convergence and guaranteed fair comparison between models.

C. Model Architecture

The proposed deepfake detection architecture will be able to process and classify video as real or fake with advanced deep learning models. The system is separated into the Preprocessing, Model Training and Evaluation and Testing phases. Training flow the model is trained on the preprocessed dataset that has been frame extracted, face cropped, normalized and resized. Some of the deep learning architectures that we used are ResNet50, VGG16, VGG19, Xception, InceptionV3, MobileNetV1, and MobileNetV2, which we used to extract features. MobileNetV1 turned out to be the most successful after training with a high accuracy, precision, recall, and AUC scores. MobileNetV1 model employs depthwise separable convolution, which is used to process features efficiently after which fully connected layer is used to make the classification. The model separates actual and fake videos according to the facial features that were obtained on the basis of the frames. The accuracy, precision, recall, ROC AUC, and PR AUC measures are used to assess the performance of the model, and a confusion matrix is created to evaluate the performance of the classification and the Table II demonstrates that the training process.

D. Training Configuration

The deep learning architectures were also trained in Jupyter Notebook on a local machine with the NVIDIA GeForce RTX 3060 (12 GB), Intel Core i7 (12th Gen) processor, and 64 GB of RAM. The binary classification between real and fake frames was trained on ImageNet-pretrained weights on the Celeb-DF v2 dataset. Input images were normalized and 224 x 224 x 3 images were resized. The model has been trained on 50 epochs, which offers the trade-off of performance and training time because additional epochs gained fewer benefits, but more computationally. Less epochs caused underfitting and more resulted in overfitting. The batch size of 16 was selected so that training can be as fast as possible with minimal memory usage and not to use high-level memory and slowness of training. Accordingly, the batch size of 16 and 50 epochs were considered the best to stabilize performance and effectively compute.

TABLE II. HYPERPARAMETER SETTINGS

No.	Variable Name	Value
1	Input size	224 × 224 × 3
2	Batch size (train/val/test)	16
3	Epochs	50, with EarlyStopping
	EarlyStopping	Monitor=val accuracy; patience=15–20; restore best weights
4	Optimizer	Adam
5	Learning Rate	0.001
6	Data Augmentation	Rotation 18°, width/height shift 0.1, shear 0.1, zoom 0.1, horizontal flip, brightness [0.9–1.1], channel shift 12
7	Epochs Fine-tuning	20
8	Activation Function	ReLU
9	Callbacks	EarlyStopping, ReduceLROnPlateau, ModelCheckpoint, LearningRateScheduler

III. RESULT AND DISCUSSION

In this section, we first discuss the performance of the proposed deepfake detection method and provide a comprehensive analysis of the obtained results. The experiments were carried out on the Celeb-DF v2 dataset, which is made up of high quality real and fake videos purposely to test the strength of the detection algorithms. In order to make a fair comparison, seven popular convolutional neural network (CNN) models, i.e. ResNet50, VGG16, VGG19, Xception, InceptionV3, MobileNetV2, and MobileNetV1, were trained and tested with the same experimental condition. Various measures of performance were used in the evaluation, such as accuracy, precision, recall, loss, and Area Under the Curve (AUC), to present all the models in a comparison manner and these results are summarized in the following sections starting with a performance table, then discussions on training behavior in terms of accuracy and loss curves, the classification reliability in terms of confusion matrices and the discriminative ability in terms of ROC and precision-recall curves. Lastly, a comparative analysis is used to point out the most effective models and their implication to deepfake detection.

Table III summarizes the performance of all the models and shows the most important model evaluation techniques, including accuracy, precision, recall, loss, ROC-AUC, and PR-AUC. As presented, MobileNetV1 was the most accurate with 0.9824 followed by ResNet50 at 0.9810. MobileNetV1 was also a little bit better than ResNet50 in the loss (0.0535 vs 0.0579) and also had a score of 1.000 on MobileNetV1 ROC-AUC and PR-AUC. These findings also show that the MobileNetV1 are very effective in discriminating between real and fake content in the Celeb-DF v2 dataset. The accuracy of Xception was next, 0.9780 then VGG19 at 0.9736. Nonetheless, VGG16 had the poorest performance as the accuracy stood at 0.9439 and loss at 0.2099, which implies that the simpler architecture of VGG16 was not coping with the difficulty of detecting deepfake.

TABLE III. PERFORMANCE METRICS OF CNN MODELS

Model	Acc	Prec	Recall	Loss	ROC AUC	PR AUC
ResNet50	0.9810	0.9810	0.9810	0.0579	0.9991	0.9992
VGG16	0.9439	0.9472	0.9439	0.2099	0.9936	0.9937
VGG19	0.9736	0.9739	0.9736	0.1588	0.9961	0.9958
Xception	0.9780	0.9788	0.9780	0.0678	0.9997	0.9997
Inception V3	0.9665	0.9667	0.9665	0.1396	0.9969	0.9969
MobileNetV2	0.9632	0.9643	0.9632	0.1182	0.9992	0.9992
MobileNetV1	0.9824	0.9824	0.9824	0.0535	0.9997	0.9997

Fig. 3 shows the training and validation accuracy curve of MobileNetV1, with a rapid and steady improvement with the 50 epochs of training. At the end, the model had an accuracy of 98% and the validation accuracy was close to the training accuracy. These minimal variations during the initial stages show effective learning and the model learns rapidly the task. The performance kept increasing with the training and it was observed that MobileNetV1 was successfully learning to differentiate between fake and original content. The fact that the training curve and the validation curve were similar

indicates that the model was not overfitting but rather it generalized. In order to address overfitting, the concept of transfer learning was utilized, in which the pre-trained weights of ImageNet were used to take advantage of features that had been previously learned, and early stopping was used to terminate training once the validation performance ceased to improve and stopped additional overfitting. This stable performance demonstrates that MobileNetV1 can be applied to more complex deepfake detection tasks with a high level of accuracy.

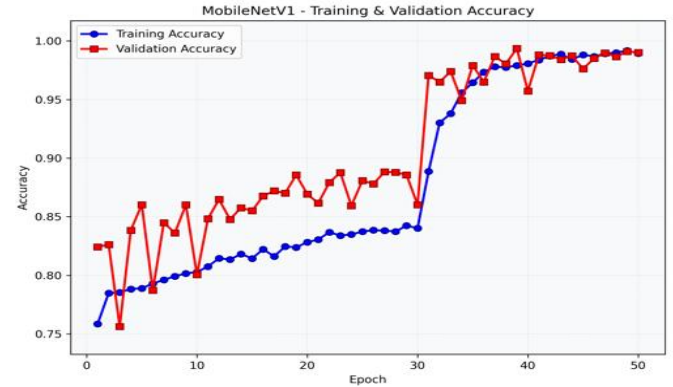


Fig. 3. Accuracy curve

Fig 4, which shows the training and validation loss curves of MobileNetV1, depicts a steady and gradual reduction in the loss during the 50 epochs. Firstly, we can observe a significant decrease in training and validation losses during the beginning of the training, as well as a steep decline during the early epochs. Following this initial downward trend, the losses level off and steadily reduce, which means that the model is actually learning and gaining the ability to differentiate between real and fake content. The fact that the loss of training (blue line) and validation loss (red line) follow a similar pattern during the training process implies that the model is not underfitting since both training and validation loss converge at the same rate. This steady decrease in loss is an indication of the MobileNetV1 model being efficient in extracting complexity patterns of the data without overfitting or underfitting. The stabilization of the two loss curves means that the model has hit a convergence point and any further training would not achieve a major improvement in the performance of the model.

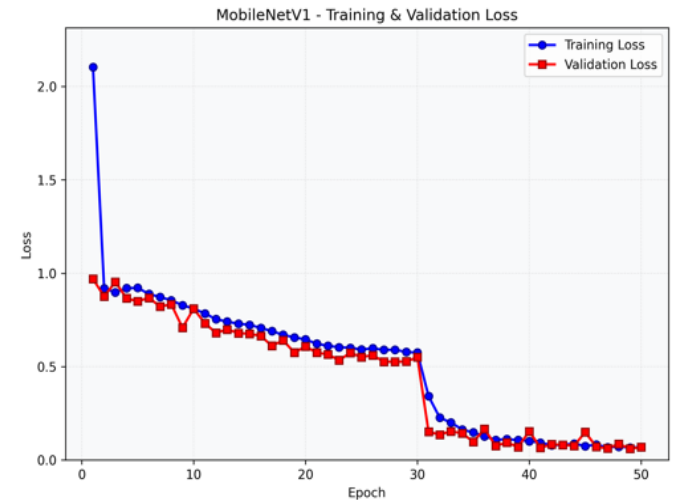


Fig. 4. Loss curve

The confusion matrix of MobileNetV1 shows in the Fig 5. that the model could differentiate between real and fake content. The matrix shows 1678 fake true positives which indicate 98.99 percent and 1699 real true positives which indicate 99.47 percent of true positives in identifying the fake and real respectively. The false positive value is very small, and only 17 fake (1.00) get incorrectly documented as real, and the false negative value is also low because 9 real (0.53) are misrepresented to be fake. These outcomes suggest that MobileNetV1 is very reliable and accurate and has little misclassification rates thus confirming its good performance in detecting deepfakes.

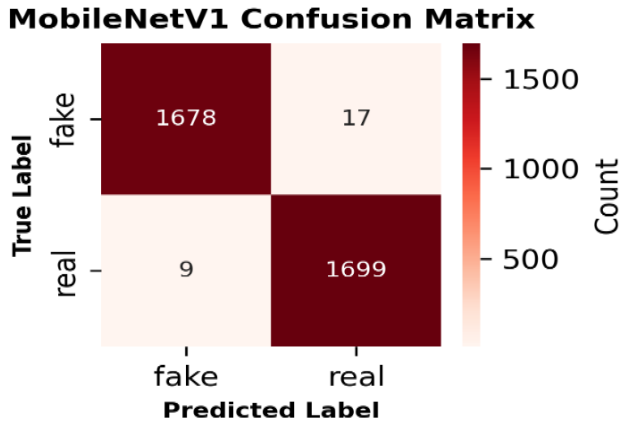


Fig. 5. Confusion Matrix

Fig 6. shows that the ROC curve of MobileNetV1, which illustrates the outstanding performance of the model in differentiating between the real and fake content. The real and fake class curve (blue and red respectively) both tend towards the top left area, and both curves signify near ideal classification. The score of both ROC-AUC is 1.000, indicating that the model does not discriminate between real and fake frames at all the thresholds. The sharp increase in the curve at the top-left corner indicates that MobileNetV1 is highly sensitive and specific in the classification of real and fake cases having fewer false positives and false negatives. Such an ideal ROC-AUC score is one more testament to the fact that the model is robust in deepfake detection.

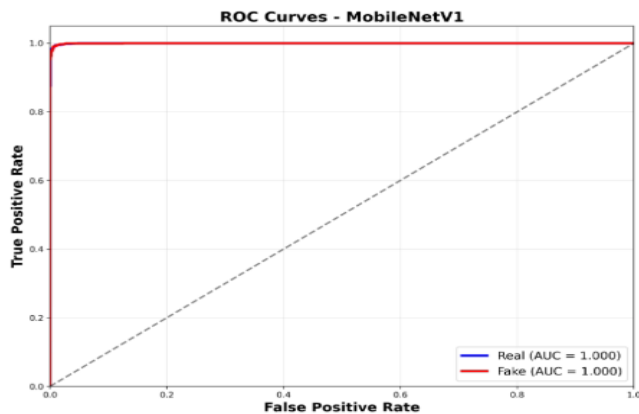


Fig. 6. ROC curve

Fig 7. shows the detailed comparison of the performance metrics of the seven CNN models ResNet50, VGG16, VGG19, Xception, InceptionV3, MobileNetV2 and MobileNetV1 and their results in terms of key metrics like accuracy, precision, recall, loss, ROC-AUC, and PR-AUC.

MobileNetV1 and ResNet50 were the best among them: the former had the highest possible accuracy of 0.9824, the latter has ROC-AUC and PR-AUC of 1.000, which is outstanding in classifying real and fake content. The next network

ResNet50, was also performing well in all benchmarks but with a very similar value of loss of 0.0579, lower than that of MobileNetV1 of 0.0535. This shows that MobileNetV1 did not only do better in leeway of accuracy but also showed good efficiency in reducing error throughout training. Conversely, VGG16 had the poorest performance as there was a greater loss of 0.2099 and an accuracy of 0.9439, which implies that the simpler structure does not effectively transfer the complexity of deepfake detection. Although Xception and VGG19 also performed well, they were a little bit worse than the MobileNetV1 and ResNet50 in accuracy and loss, which shows that the two leading models are better at their deepfake detection task.

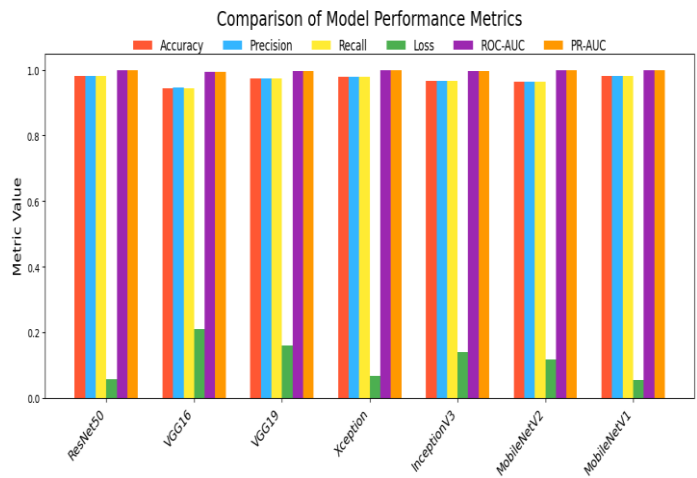


Fig. 7. Comparison of performance metrics CNN models

TABLE IV. COMPARISON OF STATE-OF-THE-ART DEEPFAKE DETECTION METHODS ON CELEB-DF DATASET

Ref	Methodology	Accur	Limitation
[12]	Modified XceptionNet transfer learning	83.7%	Lower accuracy, struggles with detecting complex deepfakes in varying conditions or low-quality images.
[13]	Masked Autoencoder-based Deepfake Detection Model (MADDMM)	86.3%	Struggles with non-facial deepfakes and may be resource-intensive during training and inference.
[5]	Depthwise separable convolution (DeepGuardNet) for effective deepfake feature extraction	91%	Higher computational cost than lightweight models, limiting real-time deployment.
[9]	XceptionNet + Self-Correcting	92.6%	Slower inference time compared to more efficient models; requires large datasets for training.
[14]	Hybrid model with Customized CNN + Ensemble Learning + Meta-Learning	96.3%	Increased complexity due to ensemble methods and meta-learning, leading to longer training and inference times.
[15]	ResNext50 and BiLSTM for deepfake detection using facial landmarks	96.5%	Limited generalization due to reliance on facial landmarks, fails with non-facial deepfakes.

[16]	Stacking DenseNet, EfficientNet-B0, Xception models with feature selection using KNN	97.7%	High memory usage and computational cost make it impractical for real-time deepfake detection.
[7]	BEiT self-supervised model with Hierarchical PatchReducer	97.5%	Long training times and requires significant computational resources, making it less suitable for real-time applications.
[17]	hard voting ensemble (XceptionNet + InceptionNet)	98.1%	Complex ensemble methods result in slower inference and high resource consumption.
Our Work	MobileNetV1	98.2%	Lightweight and efficient with high accuracy, but needs improvement in accuracy for complex deepfakes. Also, requires Explainable AI (XAI) for better transparency and trust.

In TABLE IV, it is demonstrated that the proposed deepfake detection model based on MobileNetV1 performs better than a number of state-of-the-art approaches on the Celeb-DF v2 dataset. Existing models had an accuracy of 83.75% to 97.7% with the MobileNetV1 achieve 98.24%, and a new high in detecting deepfakes. This proves that MobileNetV1 is a lightweight and efficient model, making it suitable for real-time deepfake detection, especially in resource-constrained environments.

Despite the strong performance, there are some limitations. The model is primarily trained and tested on a single dataset, which restricts its ability to generalize across different deepfake. Additionally, the model may struggle with more sophisticated deepfakes that involve compression artifacts. The lack of explainability also points to the need for Explainable AI (XAI) to improve model transparency and trust. Future work will focus on cross-dataset generalization, enhancing adversarial robustness, and extending the model to video-level detection using temporal cues. Moreover, addressing the model's interpretability through XAI techniques will be essential for broader adoption in real-world applications.

IV. CONCLUSION

This paper has provided a critical analysis of deepfake detection models on the Celeb-DF v2 dataset and compared the accuracy of seven commonly known CNN models, including ResNet50, VGG16, VGG19, Xception, InceptionV3, MobileNetV2 and MobileNetV1. The outcomes showed that MobileNetV1 and ResNet50 were the most effective as the former reached an accuracy of 0.9824, and all ROC-AUC and PR-AUC scores were equal to 1.000, which indicates its effectiveness in deepfake detection. In addition, MobileNetV1 performed better in loss than ResNet50, which demonstrates that the MobileNetV1 can achieve a high performance with the ability to reduce error. Although other models like Xception and VGG19 performed well, they were ranked at the bottom compared to the top two models particularly in accuracy and loss.

REFERENCES

[1] K. S. Kaswan, K. Malik, J. S. Dhatteval, M. S. Naruka, and D. Govardhan, "Deepfakes: A Review on Technologies, Applications and Strategies," in *International Conference on Power Energy,*

Environment and Intelligent Control, PEEIC 2023, Institute of Electrical and Electronics Engineers Inc., 2023, pp. 292–297. doi: 10.1109/PEEIC59336.2023.10450604.

[2] A. K. Singh, "Deep Learning for Deepfakes Creation and Detection", 2024. doi: 10.1201/9781003231493-3.

[3] S. Dasgupta, K. Badal, S. Chittam, M. Tasnim Alam, and K. Roy, "Attention-Enhanced CNN for High-Performance Deepfake Detection: A Multi-Dataset Study," *IEEE Access*, vol. 13, pp. 101980–101993, 2025, doi: 10.1109/ACCESS.2025.3578343.

[4] H. Singh, R. Kumar, M. Gupta, and N. Y. Joshi, "Enhanced Deepfake Detection with LSTM and ResNeXt Integration," in *3rd International Conference on Disruptive Technologies*, Institute of Electrical and Electronics Engineers Inc., pp.783–787, 2025. doi: 10.1109/ICDT63985.2025.10986702.

[5] A. N. Devi and P. Simon, "DeepGuardNet: A Novel CNN Architecture for DeepFake Image Detection," in *Procedia Computer Science*, Elsevier B.V., pp. 811–818, 2025. doi: 10.1016/j.procs.2025.04.313.

[6] H. Ranka *et al.*, "Examining the Implications of Deepfakes for Election Integrity", 2024. doi: 10.48550/arXiv.2406.14290.

[7] A. Al Redhaei, S. Fraihat, and M. A. Al-Betar, "A self-supervised BEiT model with a novel hierarchical patchReducer for efficient facial deepfake detection," *Artif Intell Rev*, vol. 58, no. 9, Sep. 2025, doi: 10.1007/s10462-025-11286-8.

[8] K. E. K. Blitti, F. G. Tola, R. Sunil, and A. Diwan, "Enhancing Deepfake Detection with Ensemble Learning: A Study on Celeb DF Dataset," in *IEEE Region 10 Annual International Conference, Proceedings/TENCON*, Institute of Electrical and Electronics Engineers Inc., pp. 319–322, 2024. doi: 10.1109/TENCON61640.2024.10903037.

[9] T. Zheng, P. Liu, and M. Dong, "A Deepfake Detection Model Based on Xception and Self-Correcting Convolutional Networks," in *2025 4th International Symposium on Computer Applications and Information Technology, ISCAIT*, Institute of Electrical and Electronics Engineers Inc., pp. 589–595, 2025. doi: 10.1109/ISCAIT64916.2025.11010714.

[10] Y. Li, X. Yang, P. Sun, H. Qi, and S. Lyu, "Celeb-DF: A Large-scale Challenging Dataset for DeepFake Forensics", 2020. doi: 10.1109/cvpr42600.2020.00327.

[11] Y. Li, D. Zhu, X. Cui, and S. Lyu, "Celeb-DF++: A Large-scale Challenging Video DeepFake Benchmark for Generalizable Forensics," Jul. 2025, doi: 10.1109/cvpr42600.2020.00327.

[12] I. Kusnadi and A. Setyanto, "Fake Video Detection using Modified XceptionNet," in *4th International Conference on Information and Communications Technology: The Role of AI in Health and Social Revolution in Turbulence Era*, Institute of Electrical and Electronics Engineers Inc., pp. 104–107, 2021. doi: 10.1109/ICOIACT53268.2021.9563923.

[13] V. K. Sharma and S. Rawat, "Enhancing Deepfake Detection Through Dynamics of Facial Expressions," Institute of Electrical and Electronics Engineers (IEEE), pp. 70–79, Jul. 2025. doi: 10.1109/iciiv64824.2025.11085942.

[14] P. Singh, S. Pandey, S. Sharma, P. Tomer, A. Bajpai, and A. Jamshed, "Hybrid Deepfake Detection Framework: Integrating Customized CNN, Ensemble Learning, and Meta-Learning for Enhanced Robustness," in *ICDT 2025 - 3rd International Conference on Disruptive Technologies*, Institute of Electrical and Electronics Engineers Inc., pp. 732–735, 2025. doi: 10.1109/ICDT63985.2025.10986628.

[15] A. A. M. Alrawahneh, S. N. H. S. Abdullah, T. Abuain, S. N. A. S. Abdullah, S. Khadijah Taylor, and N. H. S. Suhaimi, "Decision-Aid Framework for Face Authentication Detection Using ResNext50 and BiLSTM to Enhance Media Integrity," *IEEE Access*, vol. 13, pp. 89858–89873, 2025. doi: 10.1109/ACCESS.2025.3569792.

[16] S. Mohiuddin, A. Roy, S. Pani, S. Malakar, and R. Sarkar, "A feature selection-aided deep learning based deepfake video detection method," *Multimed Tools Appl*, 2025. doi: 10.1007/s11042-025-20877-0.

[17] J. Kesarwani, H. Rai, and M. Shivendra Sinha, "Ensemble Model for Image Forgery Detection Using Histogram, DWT, and DCT Feature Extraction." doi: 10.2139/ssrn.5353532.